

推导完全分类

我会把这个问题裁成“定义层裁决 + 数学证明 + 修复规格”。重点是判定：全历史一类为 0 是合法市场事实，还是趋势/中枢定义实现错误导致的结构性假零。

According to a document from 2026-07-03, this is not a “market fact” result. The all-history first-class buy/sell point count is zero because the implementation applies the trend test to the **entire accumulated center chain from data start**, not to the **current completed trend-type segment** after 走势分解; the report also shows that once any early adjacent-center relation is not same-direction, `trend_class` becomes permanently `Degenerate`, while a cut window can produce valid sell1 signals at the 2021 BTC top.

The strict ruling is:

全历史所有级别一类买卖点 = 0，是实现定义错误，不是市场事实。

More precisely:

当前实现把“某完成走势类型内的趋势”错误替换成了“全历史中枢链全同向”。

That replacement is mathematically too strong, path-start dependent, and incompatible with “走势可以分解为盘整、上涨、下跌三种走势类型的连接”的语义. The report itself states that `trend_class` is run once on the full center sequence for a level, there is no mechanism to split the center chain into multiple move types, and the code can output at most one move type for the whole level history.

1. 严格定义：第一类买卖点的作用域

Let the same-level canonical center sequence after extension/normalization be:

$$\mathcal{C}_\ell = (C_1, C_2, \dots, C_m).$$

For adjacent canonical centers define:

$$r_i = \text{Rel}(C_i, C_{i+1}) \in \{\text{Up}, \text{Down}, \text{Overlap}\}.$$

Using GG/DD, the implementation's relation rule is:

$$\text{Up} \iff DD(C_{i+1}) > GG(C_i),$$

$$\text{Down} \iff GG(C_{i+1}) < DD(C_i),$$

else overlap/expansion. The report notes that this GG/DD relation itself is aligned with center theorem II, but only when applied to **proper canonical centers**, not unextended three-segment fragments.

A trend should be a **local completed move-type block**, not the whole chain:

$$B = [C_a, \dots, C_b], \quad b > a,$$

with:

$$r_a = r_{a+1} = \dots = r_{b-1} = \text{Up}$$

or:

$$r_a = r_{a+1} = \dots = r_{b-1} = \text{Down}.$$

So the correct trend predicate is:

$$\text{TrendBlock}(a, b, d) \iff b > a \wedge \forall i \in [a, b - 1], r_i = d.$$

By contrast, the current implementation tests:

$$\forall i \in [1, m - 1], r_i = d.$$

That is:

the entire historical center chain must be one trend.

This is not the same predicate.

2. 全历史趋势门为什么会永久锁死

The current implementation's trend predicate is:

$$\text{AllTrend}(\mathcal{C}, d) = \forall i, r_i = d.$$

Suppose at some early time there exists one pair:

$$r_j \neq d$$

for both $d = \text{Up}$ and $d = \text{Down}$; for instance $r_j = \text{Overlap}$. Then:

$$\text{AllTrend}(\mathcal{C}, \text{Up}) = 0,$$

$$\text{AllTrend}(\mathcal{C}, \text{Down}) = 0.$$

Appending future centers does not change r_j . Therefore for all future prefixes:

$$\text{AllTrend}(\mathcal{C}_{1:n}, d) = 0.$$

Hence:

$$\text{trend_dir} = \text{None}$$

forever.

This proves the absorbing-lock property:

$$\exists j, r_j \notin \{\text{all Up}, \text{all Down}\} \implies \forall t' > t, \text{trend_class} = \text{None}.$$

Your report observes exactly this: the trend gate uses the full cumulative chain, the condition is broken within the first 1–17 days, and then stays closed for 8.8 years.

This is not a "market has no type1" phenomenon. It is a theorem of the current predicate.

3. 切窗能产出一类，证明当前规则依赖观测起点

Let the full center chain be:

$$(C_1, \dots, C_m).$$

Assume there is a local subchain:

$$(C_a, \dots, C_b)$$

such that:

$$\forall i \in [a, b-1], r_i = \text{Up}.$$

Then:

$$\text{TrendBlock}(a, b, \text{Up}) = 1.$$

But if the full chain contains any earlier overlap:

$$r_j = \text{Overlap}, \quad j < a,$$

then:

$$\text{AllTrend}(1, m, \text{Up}) = 0.$$

So the same price data can satisfy:

$$\text{TrendBlock} = 1$$

and:

$$\text{AllTrend} = 0.$$

If you cut the data window so that the subchain becomes the whole chain, the implementation suddenly returns Trend. Your report confirms this empirically: full history returns zero, while the 2020-10-01 to 2021-04-15 window has L4 trend up, passes all seven funnel stages, and emits two sell1 points near the 2021 BTC top.

Therefore:

当前一类触达依赖数据起点。

A structural buy/sell point detector must not depend on arbitrary file start date in this way. That is a decisive sign of a wrong trend-domain definition.

4. 中枢延伸缺失是独立第二 bug

The report states the center builder currently creates a new center every time three segments overlap, advances by 3, and explicitly leaves "extension center full start/finish interval recognition" to later; it also quotes center theorem I as saying that when subsequent intervals touch $[ZD, ZG]$, this is center extension.

So the implementation currently does:

$$u_i, u_{i+1}, u_{i+2} \Rightarrow C_1,$$

$$u_{i+3}, u_{i+4}, u_{i+5} \Rightarrow C_2,$$

$$u_{i+6}, u_{i+7}, u_{i+8} \Rightarrow C_3.$$

But if all these later submoves continue to overlap the original center region, the canonical Chanlun object should be:

one extended center C ,

not:

C_1, C_2, C_3 as same-level adjacent centers.

Formally, if a current center has core interval:

$$[ZD, ZG],$$

and the next submove has fluctuation interval:

$$[d_n, g_n],$$

then extension condition is:

$$[d_n, g_n] \cap [ZD, ZG] \neq \emptyset.$$

If true, the next submove belongs to the same extended center.

Only when:

$$d_n > ZG \quad \text{or} \quad g_n < ZD$$

does the current center fail to extend and a new move/center relation become possible.

The report observes that L0 all-history adjacent center relations are 94.4% overlapping, precisely because extension is not absorbed; those overlaps then feed

`LevelExpansion` into the same-level chain and make "all same direction" impossible.

Thus even after fixing local trend blocks, if center extension is not implemented, the center chain is still polluted by same-level fragments.

5. 正确的 canonical 中枢链构造

Define a canonical center as:

$$C = (start, end, ZD, ZG, DD, GG, submoves).$$

Given consecutive lower-level moves $u_1, u_2, \dots$:

Step 1: seed center

Find the first triple u_i, u_{i+1}, u_{i+2} whose price ranges overlap enough to form:

$$[ZD, ZG] \neq \emptyset.$$

Create center:

$$C = [u_i, u_{i+2}].$$

Step 2: extension

For next lower-level move u_j , compute its interval:

$$[d_j, g_j].$$

If:

$$[d_j, g_j] \cap [ZD, ZG] \neq \emptyset,$$

then:

$$C.end := u_j.end,$$

and update outer bounds:

$$DD := \min(DD, d_j), \quad GG := \max(GG, g_j).$$

Whether ZD/ZG core is frozen or updated depends on your exact center convention, but the key is:

u_j remains part of the same center, not a new same-level center.

Step 3: non-extension

If:

$$d_j > ZG \quad \text{or} \quad g_j < ZD,$$

then the center stops extending. The following structure becomes either:

1. a same-level trend continuation center,
2. a higher-level center,
3. or a new move-type segment.

Only after this canonicalization should you apply center relation:

$$DD(C_{i+1}) > GG(C_i)$$

or:

$$GG(C_{i+1}) < DD(C_i).$$

So the relation rule is not wrong; its input object is wrong.

6. 正确走势类型分解算法

Once centers are canonical, parse the center chain into move-type blocks.

Let:

$$R_i = \text{Rel}(C_i, C_{i+1}).$$

Define a maximal same-direction trend block:

$$B = [C_a, \dots, C_b]$$

such that:

$$b > a$$

and:

$$R_a = R_{a+1} = \dots = R_{b-1} = d \in \{\text{Up}, \text{Down}\}.$$

Maximal means it cannot extend left or right without breaking same-direction relation.

A consolidation block is either:

1. one canonical center without a same-direction successor, or
2. a sequence whose adjacent relation is overlap/expansion and should be absorbed into a higher-level center or treated as consolidation.

Then define:

$$MoveType(B) = \begin{cases} Trend(d), & B \text{ has at least two centers same direction,} \\ Consolidation, & B \text{ is one center or non-trend extension structure.} \end{cases}$$

This gives a decomposition:

$$C_\ell = B_1 \oplus B_2 \oplus \dots \oplus B_k.$$

Each B_j is one completed or currently active 走势类型.

Now the type1 trend gate must be applied to the current/last completed B_j , not to all centers:

$$trend_dir(B_j) = d$$

if:

$$MoveType(B_j) = Trend(d).$$

This is the mathematical repair for the trend gate.

7. 第一类买卖点的正确判据

For a trend block:

$$B = [C_a, \dots, C_b], \quad MoveType(B) = Trend(d),$$

with $b > a$, define:

$$C_{last} = C_b, \quad C_{prev} = C_{b-1}.$$

The type1 candidate segment must be the outgoing lower-level move C -segment from the last center in direction d . It must break the last center:

For downward trend:

$$d = \text{Down}, \quad \min P(C) < ZD(C_{last}).$$

For upward trend:

$$d = \text{Up}, \quad \max P(C) > ZG(C_{last}).$$

Then compare it to the previous comparable A -segment, which is not a single bar/segment unless the lower-level move type is actually single-segment. It is the lower-level completed move type that leaves the previous center in the same direction.

So the correct A/C objects are:

$A = \text{lower-level move type leaving } C_{prev},$

$C = \text{lower-level move type leaving } C_{last}.$

The report notes that current code uses a single segment for A and C, while the cited A/B/C description speaks of "C segment's move type completed," so this is another under-implementation after the trend gate is fixed.

Back divergence is:

$$Weak(C, A) = 1,$$

where *Weak* should be defined by a force measure, not only MACD area.

Then:

$$d = \text{Down} \Rightarrow B1.$$

$$d = \text{Up} \Rightarrow S1.$$

8. 对 8 个问题逐条裁决

Q1. 趋势定义的作用域

裁决：

必须作用于走势分解后的当前走势类型内的中枢子链。

不能作用于自数据起点累计的全历史中枢链。

理由：

1. 趋势定义的对象是“某完成走势类型”；
2. 走势分解定理说历史是盘整、上涨、下跌三种走势类型的连接；
3. 若趋势门作用于全历史，等价于要求“全历史是一段趋势”，这与分解定理冲突；
4. 实测显示它导致早期锁死，并且一类是否出现取决于观测窗起点，而不是结构本身。

正确算法

1. 先 canonicalize centers with extension.
2. Compute adjacent center relations.
3. Segment the relation chain into maximal trend/consolidation blocks.
4. Apply type1 gate only within the active or just-completed trend block.
5. “最后一个中枢” means last center of that trend block, not latest center in global history.

Q2. 中枢延伸 vs 新建

裁决：

连续围绕同一区间震荡，应记为一个延伸中枢，不应拆成多个同级别中枢。

If nine lower-level moves keep touching the same $[ZD, ZG]$, the correct object is one extended center with longer duration.

If the engine creates three same-level centers and they overlap, those overlap relations should not remain in the same-level chain for trend continuation tests. They either belong to the same extended center or imply formation of a higher-level center. Your report's code fact and theorem comparison say exactly this.

Therefore:

\boxed{\text{当前 94.4\% overlapping center chain 是中心延伸缺失的实现产物。}}

Q3. “依次同向”的 GG/DD 还是 ZG/ZD?

裁决:

趋势延续的强标准是 GG/DD 完全分离。

The implementation's relation rule:

$$DD_{next} > GG_{prev}$$

for upward continuation and:

$$GG_{next} < DD_{prev}$$

for downward continuation is faithful to the cited center theorem II. The report says the issue is not the GG/DD relation itself, but that it is applied to unextended fragment centers.

So the ruling is:

GG/DD 口径保留；先修中枢规范化。

Using ZD/ZG non-overlap would be a weaker relation and may detect looser separation, but it is not the same as the strict continuation theorem. If you implement it, label it as a relaxed variant, not canonical trend continuation.

Q4. 盘整背驰与第一类

裁决:

标准第一类买卖点应锚定趋势背驰；

but:

盘整背驰不能消失；它必须被某级别买卖点或小转大/区间套证书承接。

The current implementation:

$$\tau = Consolidation \Rightarrow trend_dir = None \Rightarrow no\ B1/S1$$

is acceptable only if another channel records the resulting lower-level or higher-level buy/sell point. But the report says consolidation divergence currently has no independent signal channel.

So the full ruling is:

- Do not call every consolidation divergence same-level B1/S1.
- But do not drop it.
- Route it through:

$$N_{\ell \downarrow e}^{\delta}$$

or a “pan-divergence / small-to-large” certificate:

$$PanDiv_{\ell}^{\delta} \Rightarrow \exists e < \ell, Conf_e^{\delta}$$

or:

$$PanDiv_{\ell}^{\delta} \Rightarrow XZD_{\ell \downarrow e}^{\delta}.$$

Thus current strict trend-only type1 is incomplete if no other channel marks consolidation endings.

Q5. A/C 段对象粒度

裁决:

A/C 应为次级别走势类型，不应默认是单一线段。

If a lower-level departure move consists of multiple segments, the MACD/force comparison should cover the whole lower-level move type interval:

$$I(A) = [\lambda_A, \rho_A], \quad I(C) = [\lambda_C, \rho_C].$$

Then:

$$Area(A) = \sum_{t \in I(A)} |hist_t|,$$

$$Area(C) = \sum_{t \in I(C)} |hist_t|.$$

Current code chooses one segment for A and one segment for C; the report explicitly identifies this mismatch.

So after fixing trend decomposition, also fix:

$$A, C : \text{Segment} \rightarrow \text{LowerLevelMoveType}.$$

Q6. 全历史所有级别 0 个一类是否可能是市场事实?

裁决:

在当前数据与当前漏斗证据下，它不是市场事实，而是实现错误。

Strictly, an arbitrary finite price path could in theory have no trend divergence under a very strict definition. But your actual observed zero has a sufficient implementation cause:

1. The trend gate uses the whole cumulative chain.
2. The first early non-same relation locks it forever.
3. Downstream type1 logic is never evaluated.
4. A cut window with the same code produces structurally valid sell1 points near a historical top.

Therefore this zero is not evidence that BTC had no type1 points. It is evidence that:

the implemented trend gate has the wrong domain.

So:

“几何环瓶颈”和“市场事实”归因均被本漏斗推翻。

The report's result package says this explicitly.

Q7. 高级别线段方向如何确定?

裁决:

高级别“线段”方向应来自其所代表的次级别走势类型方向，而不是相邻单元 hi 比较占位。

If the higher-level unit is a lower-level completed trend:

$$MoveType = Trend(Up) \Rightarrow Direction = Up,$$

$$MoveType = Trend(Down) \Rightarrow Direction = Down.$$

If the lower-level completed move is consolidation:

$$MoveType = Consolidation,$$

then it is not a directional departure leg for type1 unless it is further resolved through small-to-large or interval-nesting context.

Endpoint comparison can be used as fallback geometry, but it should not be the canonical direction for type1 break logic.

The report notes that current higher-level direction is a placeholder based on exterior high comparison and has not been tested in mixed-chain type1 because the trend gate blocks everything.

Q8. 流式下“最后一个中枢”的边界

裁决：

“最后一个中枢”必须相对于当前走势类型，而不是全历史最近中枢。

Define current trend context at time t :

$$B_t = [C_a, \dots, C_b].$$

Then:

$$LastCenter_t = C_b,$$

$$PrevCenter_t = C_{b-1}$$

provided:

$$b > a.$$

The current trend start a is determined by the last completed move-type boundary. A new boundary opens when:

1. the current center stops extending;
2. a new canonical center appears outside the extension relation;
3. the relation sequence changes from same-direction continuation to overlap/opposite relation;
4. or a higher-level center/consolidation is formed.

So in streaming mode maintain:

$$CurrentMove_t = (start_center, end_center, move_type, dir, status).$$

Then “last center” is meaningful.

Without this, “last center” degenerates into:

latest center in entire history,

which is exactly the wrong behavior described in the report.

9. 修复后的数学规格

The corrected pipeline should be:

raw lower-level moves $\rightarrow$ canonical centers with extension $\rightarrow$ center relation chain $\rightarrow$ move-type decomposition

Formal functions:

9.1 Center extension

$$Centers_\ell = ExtendCenters(UnitRanges_{\ell-1}).$$

9.2 Move decomposition

$$Moves_\ell = Decompose(Centers_\ell).$$

Each move:

$$M = (C_a, \dots, C_b, kind, dir, status).$$

9.3 Type1 candidate

For move M :

$$Type1Cand(M, s)$$

iff:

1. $M.kind = Trend(d)$;
2. M has at least two centers;
3. s is the lower-level move type leaving the last center in direction d ;
4. s breaks the last center boundary;
5. $Weak(s, A) = 1$, where A is the previous comparable lower-level departure move.

Then:

$$\begin{aligned} d = Down &\Rightarrow B1, \\ d = Up &\Rightarrow S1. \end{aligned}$$

10. 严格证明修复后不会全局锁死

Under the corrected trend predicate, a local block:

$$B = [C_a, \dots, C_b]$$

with all relations Up defines:

$$TrendBlock(B, Up) = 1$$

regardless of earlier relation r_j for $j < a$.

So if the history contains an early overlap but later contains a local up trend:

$$r_j = Overlap, \quad r_a = \dots = r_{b-1} = Up,$$

then:

$$AllTrend(C) = 0,$$

but:

$$TrendBlock(a, b, Up) = 1.$$

Thus the corrected predicate detects local trends and is not permanently locked by early history.

This matches the intended走势分解 semantics.

11. 最终裁定

一类零触达不是市场事实。

根因是两个定义级实现错误：

1. `trend_class` applies to the entire historical center chain instead of current completed move-type block.
2. center extension is missing, so one extended center is split into many overlapping same-level fragments.

A/C single-segment granularity, consolidation divergence routing, higher-level direction placeholder, and MACD-only force are secondary defects that become relevant after the trend gate is repaired.

The correct repair order is:

1. center extension → 2. move-type decomposition → 3. local trend gate → 4. A/C as lower-level move

After that, all previous "type1 = 0" downstream alpha conclusions must be rerun.

Until then:

the engine has not actually tested Chanlun first-class buy/sell points.

